

ECONOMY — Agriculture and Allied Sector

Batch 8 of 10 · Source: Lucent §4 "Agriculture and Allied Sector" only — Parmar's Economics section has no dedicated agriculture chapter · PYQ bucket "09 Agriculture and Food Management" (24 Qs) · ★ = tested in a real past SSC paper

Green Revolution / HYV / agriculture content also lives in **claudefgeography-raw/agriculture_raw_extraction.md** (Geography build) — cross-referenced, not duplicated in full.

1 Green Revolution — Reach & Regional Imbalance

HOOK Green Revolution (1960s) — HYV seeds, fertilizers, irrigation → self-sufficiency in foodgrains. **Only effective where irrigation was already good** ★ — Punjab, Haryana, western UP ★ (canal/tube-well irrigation, wheat-centric farming); largely **ineffective in Bihar** in its first phase ★ (flood-prone, weak infra, small landholdings, limited irrigation). **Punjab = first & largest beneficiary** ★.

TRAP Regional disparity cause ★ = HYV/irrigation access **restricted to select states** ★ — NOT price controls or fertilizer neglect (tested distractors). A tested Assertion-Reason confirms BOTH true, R correctly explains A ★ (regional imbalance ← HYV/irrigation initially available only in select states).

Green Revolution did **NOT** cover barley, ragi, minor millets — confined to **rice, wheat, maize, jowar**; biggest effect on **wheat** (~500% increase, per raw extraction). Credited internationally to **Norman Borlaug** ("Father of Green Revolution"); in India, **Dr. M.S. Swaminathan** played the leading role.

DOUBT Raw extraction's own internal inconsistency: Lucent p.257 says Green Revolution began under a "termed name" credited to "William Gande" (likely a garbled "William Gaud", the US administrator who coined the term "Green Revolution" in 1968) alongside Borlaug/Swaminathan; p.258 separately states it "started in the Third Five Year Plan" — this conflicts with Parmar Ch.10 (Batch 7), which places it in the Plan Holiday (1966-69). Both dates transcribed as printed, not resolved.

2 Occupational Structure & Land Reforms (1947-1990)

HOOK At Independence (1947), ~**70-75%** of India's population depended on agriculture ★ (highest GDP contributor then, since manufacturing/services were nascent). By 1990, this had only fallen to ~**65%** ★ — despite agriculture's *GDP share* falling sharply, its *workforce share* barely moved: a tested "failure of structural transformation" ★ (workforce didn't shift to industry/services as theory predicts). A tested Assertion-Reason (planning 1951-90 failed to transform employment structure / >65% workforce still in agriculture) — **both true, R explains A** ★.

Occupational diversification ★ = the term for workers shifting from agriculture to industry/services (reduces rural risk, generates new income — NOT "industrialization"/"urbanization" alone, tested distractors).

HOOK Land reforms (1947-90) objective ★: **abolish intermediaries (zamindars) + land ceilings** ★ → redistribute surplus land to landless/small cultivators, reduce land-ownership concentration ★ — "land to the tiller" principle (NOT consolidate land in a few hands, promote cash crops, or mechanisation — tested distractors).

Land measure	Meaning
Land Ceiling	Maximum land a farmer may legally hold
Chakbandi (Consolidation)	Aggregating fragmented/divided landholdings — first implemented 1920, Baroda

Landholding size classes: Marginal <1 ha; Small 1-4 ha; Large >4 ha.

TRAP **Fertilizer subsidies (1960-90)** ★ were a "double-edged" instrument — boosted productivity **but disproportionately benefited richer/larger farmers** ★ (small farmers lacked resources to fully exploit them) — worsening rural inequality, NOT reducing employment or discouraging HYV use.

UP = highest share of workforce in agriculture ★ among the states tested (fertile Ganga-Yamuna plain, large rural population) — **Punjab/Haryana >85% of net sown area under irrigation** ★ (Bhakra Nangal canal system + tubewells).

3 The Revolutions

Revolution	Sector	Key facts ★
White Revolution ★	Milk	Operation Flood ★ — launched Jan 1970 ★ by NDDB ; father: Dr. Verghese Kurien ★ (also Amul model/Anand pattern architect — NOT the inventor of milk pasteurisation, that's Louis Pasteur, 19th century — a tested distractor ★); made India the world's largest milk producer
Yellow Revolution	Oilseeds	Named in the raw extraction ("increase in oil-seeds production"), but the Technology Mission on Oilseeds (1986)/NDDB role/"Operation Golden Flow" detail is PYQ-book-only, see A.2
Blue Revolution	Fish production	—
Sweet Revolution	Beekeeping/honey	National Beekeeping & Honey Mission (NBHM), ₹500cr, 2020-23, via National Bee Board

HOOK **Amul** ★ (Kaira District Co-op Milk Producers' Union, PYQ-book-only, see A.2) — established **14 Dec 1946** ★, Anand, Gujarat, under **Sardar Vallabhbhai Patel's** guidance — laid the foundation of the White Revolution by ending middlemen exploitation.

4 Institutions, Certification & Boards

Body	Role/Act
AGMARK ★	Quality/purity certification for agricultural products ★, under Agricultural Produce (Grading and Marking) Act, 1937
ACPC/CACP	Recommends Minimum Support Prices (MSP) — Lucent calls it "Agriculture Cost and Price Commission (ACPC)"; Parmar's Ch.10 uses "CACP" (Commission for Agricultural Costs & Prices, est. 1 Jan 1965, see Batch 7)
NABARD	Apex rural-credit institution, 12 July 1982, Sivaraman Committee (see Batch 5/7); authorised capital ₹500cr → ₹5,000cr (2001) → ₹30,000cr (2018 Amendment Bill)

Commodity Board	HQ	Governing Act
Coffee Board	Bengaluru	Coffee Act, 1942
Rubber Board	Kottayam	Rubber Act, 1947
Tea Board	Kolkata	Tea Act, 1953
Tobacco Board	Guntur	Tobacco Act (AP), 1975
Spices Board	Kochi	Spices Act, 1986

MSP = price at which govt commits to buy directly from farmers if market price falls below it.

Commercial crops = grown for trade, not self-consumption (oilseeds, sugar, fibre, narcotic, beverage crops).

5 Agricultural Finance

Unorganised sources	Organised sources
Money-lenders, relatives, landlords, commission agents	Co-op societies/banks, commercial banks, RRBs, government

Co-operative Credit first started **1904**. **Land Development (Mortgage) Bank** established **1919** — long-term loans.

6 Modern Schemes & Concepts

Scheme/Concept	Key facts ★
PMFBY	Launched 2016 Kharif season — comprehensive crop-failure insurance; farmer premium capped at 1.5%(Rabi)/2%(Kharif)
Agriculture Infrastructure Fund (AIF) ★ (PYQ-book-only)	Finances agri-logistics and cold chains ★ (warehouses, primary processing) — loans to farmers/FPOs/cooperatives (NOT free fertilizer, cash transfers, or MGNREGA — tested distractors)
National Mission on Natural Farming ★	Launched Nov 2024 , targets 1 crore farmers ★ for chemical-free/natural farming
ULPIN ★	14-digit unique land-parcel ID ★; tracks ownership/resolves disputes; covers both rural and urban land ★ (NOT urban-only — tested distractor)
SMAM ★	Sub-Mission on Agricultural Mechanization — 75% grant to FPOs ★ for Kisan Drone demonstrations
GMOs ★	Purpose: improve crop yield, pest resistance, nutritional quality ★ (e.g. Bt Cotton for pest resistance, Golden Rice for Vitamin-A) — NOT eliminate fertilizer/pesticide use entirely

7 Agriculture — Growth, Production & National Data (Lucent \$4, Full Coverage)

Indian agriculture grew at an average annual rate of 4.6% over the last 6 years (per source); 3.0% in 2021-22 vs 3.3% in 2020-21. India has rapidly become a net agri-exporter — agri/allied exports grew 18% in 2020-21; 2021-22 agri exports hit an all-time high of **US\$50.2 billion**. Agriculture & Allied = 18.4% of India's GVA at current prices (2022-23, BHARAT 2025); accounts for ~1/7th of GDP and sustains ~2/3rds of the population.

Real GVA grew 6.2% in H1 FY25 (strong Q1, subdued Q2); agriculture and services were the key growth drivers. Agri growth: Q2 FY25 = 3.5% (best in 4 quarters), aided by healthy Kharif output, above-normal monsoon, adequate reservoir levels. First advance estimates (2024-25): Kharif foodgrain production = record **1,647.05 lakh metric tonnes (LMT)**, +5.7% over 2023-24, +8.2% over the 5-year average. As of 10 Jan 2025, Rabi wheat/gram sowing was up 1.4%/0.8% YoY.

Foodgrain production rose from 265.05 million tonnes (2013-14) to a record 328.85 million tonnes (2023-24); horticulture production = 352.23 million tonnes (3rd advance estimate). (Source: BHARAT 2025)

International Year of Millets — UNGA's 75th session (March 2021) declared 2023 the International Year of Millets; millets are "Smart Food" — high nutrition, climate-resilient, aligned with UN SDGs.

8 PMFBY, MSP & Related Schemes — Full Detail

PMFBY (Pradhan Mantri Fasal Bima Yojana) — launched 2016 Kharif season for comprehensive crop-failure insurance/income stabilisation. World's **largest crop-insurance scheme by farmer enrolment** (avg. 5.5 crore applications/year) and 3rd-largest by premium received. Farmer premium share: 1.5% (Rabi) / 2% (Kharif) of total premium — Centre+State bear the rest. Over its first 6 years, farmers paid ₹25,186 crore in premiums and received ₹1.2 lakh crore in claims (as of 31 Oct 2022).

Allied sectors (horticulture, animal husbandry, dairy, fisheries) are important for rural economic conditions/nutrition. **Commercial crops** = grown for trade, not self-consumption — includes oilseed, sugar, fibre, narcotic, and beverage crops. **ACPC (Agriculture Cost and Price Commission, per Lucent's naming)** decides MSP on the government's behalf. **MSP** = the price at which government commits to purchase crops directly from farmers if the market price falls below it.

9 Sweet Revolution, Blue Economy 2.0 & Full Commodity Board Table

Sweet Revolution — govt approved ₹500 crore for the **National Beekeeping & Honey Mission (NBHM)** over 3 years (2020-21 to 2022-23), under the Atma Nirbhar Bharat (ANB) scheme; promotes scientific beekeeping via the **National Bee Board (NBB)**.

Blue Economy 2.0 — highlighted in the Interim Budget 2024 (1 Feb 2024); emphasises sustainable development of coastal aquaculture, seas, and coasts; pillars: Mariculture, Maritime Transport, Renewable Energy, Tourism; balances environmental preservation with economic activity. World Bank's definition: "sustainable use of ocean resources for economic growth, improved livelihoods and jobs while preserving the health of the ocean ecosystem."

Board	HQ	Governing Act/Date
Coffee Board	Bengaluru, Karnataka	Coffee Act, 1942
Rubber Board	Kottayam, Kerala	Rubber Act (Kerala), 1947
Tea Board	Kolkata, West Bengal	Tea Act, 1953
Tobacco Board	Guntur, Andhra Pradesh	Tobacco Act (A.P.), 1975
Spices Board	Kochi, Kerala	Spices Act, 1986
National Meat and Poultry Processing Board	Delhi	26 Dec 2008
Indian Grape Processing Board	Pune, Maharashtra	2 Jan 2009

10 Allied Sectors — Livestock, Fisheries & Dairy

Livestock, forestry/logging, and fishing/aquaculture are buoyant-growth allied sectors and a potential source of better farm incomes. **Livestock sector** grew at 7.9% CAGR (2014-15 to 2020-21, constant prices); its share of agriculture GVA rose from 24.3% (2014-15) to 30.1% (2020-21). **Fisheries sector** — ~7% annual average growth since 2016-17; ~6.7% share of agriculture GVA. Allied sectors' faster growth than crops raises their share of total agri-GVA over time.

Dairy — the most critical livestock component, employing 8+ crore farmers directly; India ranks **1st in milk production** globally, **3rd in egg production**, **8th in meat production**.

11 National Food Security Act (NFSA), 2013

Objective: food/nutritional security across the human life cycle via affordable access to adequate quality food, ensuring dignity. Notified **10 September 2013**; covers up to 75% of the rural population and up to 50% of the urban population for subsidised foodgrains under the **Targeted Public Distribution System (TPDS)** — about 2/3rds of the total population; special nutritional focus on women and children. (Source: MOCAFPD)

12 Green Revolution, Revolutions & Land Reform — Additional Lucent Detail

Per Lucent: credit for the Green Revolution in India goes to agricultural scientist **Dr. Norman Borlaug**; **Dr. M.S. Swaminathan's** contribution is "not less" — but the "termed name" is credited to American scientist "William Gande" (a likely garbled rendering of William Gaud, who coined the term "Green Revolution" in 1968). Elsewhere the book separately calls Swaminathan the leading figure in India's Green Revolution and Borlaug "The Father of Green Revolution" in the world — an internal inconsistency, both preserved verbatim.

White Revolution = rapid rise in milk production; driven by **Operation Flood**, "Father": Dr. Verghese Kurien; the world's largest integrated dairy-development programme, started by the **National Dairy Development Board in 1970**. **Yellow Revolution** = rise in oilseed production. **Blue Revolution** = rise in fish production.

11th Plan (2007-12): agriculture's plan outlay fell to just 18.5%; average annual agri/allied GDP growth = 3.7% — short of the 4% target but better than the 10th Plan's 2.4%. The **10th Plan was the first plan not to fix crop-production targets**. Green Revolution did not cover barley, ragi, or minor millets — confined to HYV rice, wheat, maize, jowar. **National Agriculture Insurance Scheme** implemented October 1999.

Land-holding reorganisation used two measures: **Land Ceiling** (max legal landholding) and **Chakbandi** (consolidating fragmented/divided land) — Chakbandi was first implemented in India in **1920, Baroda**. Landholding classes: Marginal <1 ha, Small 1-4 ha, Large >4 ha.

Per the source's own pink-box correction: **M.S. Swaminathan**, Indian geneticist/international administrator, is renowned for leading India's Green Revolution, while **Norman E. Borlaug** is called "The Father of Green Revolution" globally. Green Revolution "started in the Third Five Year Plan" (per Lucent p.258) — its most positive effect was on wheat, with a 500% increase in crop production (as printed, unverified statistic).

13 Agricultural Finance & NABARD — Full Detail

Unorganised finance sources: money-lenders, money-dealers, relatives, businessmen, landlords, commission agents. **Organised sources:** Co-operative Committees, Co-operative Banks, Commercial Banks, RRBs, the Government. Co-operative Credit Organisation started 1904; Primary Co-operative Committees give short-period credit; **State Co-operative Agriculture and Rural Development Banks** give long-period credit. **Land Development Bank** (originally "Land Mortgage Bank") established 1919 — long-term loans.

NABARD — apex rural-credit institution; established **12 July 1982** by merging the RBI's Agriculture Credit Department and the Agricultural Refinance and Development Corporation; based on the **Sivaraman Committee** (printed "Shivraman") recommendations. Authorised share capital: ₹500 crore initially → increased to **₹5,000 crore, effective 1 February 2001** → a further six-fold increase to **₹30,000 crore** via the NABARD (Amendment) Bill 2017 (passed January 2018), which also enabled RBI's exit from NABARD's ownership.

Appendix

A.1 — PYQ citation index (bucket "09 Agriculture and Food Management", 24 Qs)

Fact tested	Exam · Date · Shift
AIF supports agri-logistics/cold chains	SSC CGL Tier-I, 13/09/2025 (Sh-01)
Green Revolution regional disparity cause	SSC CGL Tier-I, 21/09/2025 (Sh-01)
Structural transformation failure (workforce share)	SSC CGL Tier-I, 22/09/2025 (Sh-01)
Verghese Kurien = Operation Flood/White Revolution	SSC CGL Tier-I, 22/09/2025 (Sh-03)
Green Revolution phase-1 ineffective in Bihar	SSC CGL Tier-I, 25/09/2025 (Sh-02)
Land reforms equity goal = abolish intermediaries	SSC CGL Tier-I, 26/09/2025 (Sh-01)
Green Revolution → regional imbalance (A-R)	SSC CPO Tier-I, 10/12/2025 (Sh-01)
1951-90 planning failed employment transformation (A-R)	SSC CPO Tier-I, 11/12/2025 (Sh-02)
AGMARK = agriculture products certification	SSC CHSL Tier-I, 21/11/2025 (Sh-03)
National Mission on Natural Farming, 1 crore farmers	SSC CHSL Tier-I, 22/11/2025 (Sh-03)
Punjab/Haryana >85% net sown area irrigated	SSC CHSL Tier-I, 22/11/2025 (Sh-03)
Operation Flood launched 1970	SSC CHSL Tier-I, 26/11/2025 (Sh-02)
NDDB's role in Technology Mission on Oilseeds	SSC Stenographer, 06/08/2025 (Sh-01)
ULPIN system statements	SSC Stenographer, 06/08/2025 (Sh-01)
UP = highest workforce share in agriculture	SSC Stenographer, 06/08/2025 (Sh-01)
Occupational diversification term	SSC Stenographer, 06/08/2025 (Sh-01)
Punjab = first Green Revolution beneficiary	SSC Stenographer, 06/08/2025 (Sh-02)
Technology Mission on Oilseeds ministry	SSC Stenographer, 06/08/2025 (Sh-02)
GMO purpose = yield/pest-resistance/nutrition	SSC Stenographer, 07/08/2025 (Sh-01)
Main occupation post-Independence = agriculture	SSC Stenographer, 07/08/2025 (Sh-03)
Land reform objective 1947-90 = reduce concentration	SSC Stenographer, 08/08/2025 (Sh-01)
Fertilizer subsidy = double-edged, favoured richer farmers	SSC CGL Tier-I, 21/09/2025 (Sh-03)
SMAM: 75% grant for Kisan Drone demos	SSC Phase XIII, 29/08/2025 (Sh-01)

A.2 — PYQ-book-only facts

The Agriculture Infrastructure Fund (AIF), the Technology Mission on Oilseeds (1986)/NDDB's role/"Operation Golden Flow", the Amul/Kaira District Co-op founding details (1946, Sardar Patel) — none of these are in Lucent §4's raw extraction (which only names Verghese Kurien/Operation Flood and "Yellow Revolution" in passing); all sourced transparently to the PYQ book's own explanation text.

A.3 — Doubts carried forward

Green Revolution's start-date attribution (Third Plan per Lucent p.258 vs Plan Holiday 1966-69 per Parmar Ch.10) — both transcribed as printed, not reconciled; "William Gande"/Gaud naming confusion in the source.

A.4 — Full source coverage note

Sections 7-13 above restore all previously-condensed Lucent §4 content in full: national agri growth/production statistics, PMFBY's complete premium/enrolment detail, Sweet Revolution/Blue Economy 2.0, the complete 7-row commodity-board table, allied-sector (livestock/fisheries/dairy) statistics, the National Food Security Act's full provisions, the raw extraction's own internally-inconsistent Green-Revolution attribution text, and NABARD's full capital-history timeline. Every fact in the raw extraction for this batch is now covered — nothing is deliberately excluded.